

Evaluating Sweet Potato as an Intervention Food to Prevent Vitamin A Deficiency

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Abstract: Vitamin A (VA) deficiency causes over 600000 deaths per year, mostly of young children or pregnant women. Populations prone to VA deficiency obtain about 82% of their VA from plant sources that are rich in pro-VA carotenoids such as beta-carotene. Orange-fleshed sweet potatoes (OFSP) are an especially good source. We evaluated OFSP carotenoid concentrations, bioaccessibility, and cooking and storage, then used this to estimate the amount of OFSP needed to supply 100% of VA for people at risk for deficiency. The grams/day of OFSP needed to meet VA requirements varies with age and sex, and with the amount of beta-carotene in the OFSP. Amounts ranged from 6 to 33 g/d (0.02 to 0.13 cups/d) for a 3-y-old child with marginal VA status; to 68 to 381 g/d, (0.27 to 1.49 cups/d) for a lactating woman with good status. These are amounts that can be eaten on a daily basis. The amount of OFSP needed to supply the VA requirement to all of the 208.1 million people most in danger of VA deficiency for 1 y is 2.1 to 11.7 million metric tons, or 2% to 11% of current world sweet potato production. The most important factor influencing the effectiveness of sweet potato for preventing VA deficiency, by far, is the variety of sweet potato used. Fat in the diet is also important. We conclude that OFSP could prevent VA deficiency in many food-deficit countries—if OFSP were substituted for white, cream, yellow, or purple sweet potatoes.

Sweet Potatoes

Sweet potato (*Ipomoea batatas*) is a perennial tuber. It is a member of the Convolvulaceae family, which also contains the morning glory. Flowers can be white or purple, and leaves can be green or purple. Flesh can be white, cream, yellow, orange, or purple (Woolfe 1992; Bovell-Benjamin 2007), with orange, white, and cream the most commonly grown and eaten (Sandhill Preservation Center 2010). Both the leaves and, more commonly, the tuberous roots are eaten (Woolfe 1992; Bovell-Benjamin 2007). Sweet potatoes grow well in tropical, subtropical, and temperate areas. Sweet potatoes originated in the New World and were introduced into Spain, India, and the Philippines by Spanish explorers in the 15th and 16th centuries. Their distribution is now worldwide. In parts of Africa, Asia, and the Pacific, sweet potatoes are an important staple crop (Woolfe 1992; Bovell-Benjamin 2007).

Sweet potatoes can be grown from seeds, but they mainly are propagated from root cuttings, a simple technique useful for subsistence farming. They grow well in hot, humid climates but need a rather long growing season of approximately 90 to 150 frost-free days (Bovell-Benjamin 2007). Sweet potatoes normally flower in summer and bear fruit in late summer and fall, thus providing a source of carotenoids and vitamin A (VA) in the fall and winter.

Sweet potatoes are a nutritious food, low in fat and protein, but rich in carbohydrate. Tubers and leaves are good sources of antioxidants (Teow and others 2007), fiber, zinc, potassium, sodium, manganese, calcium, magnesium, iron, and vitamin C (Antia and others 2006). Orange-fleshed sweet potatoes (OFSP) are also very good sources of VA (Hagenimana and others 1999a, 1999b; Teow and others 2007; Wu and others 2008). Because of their nutritional qualities, sweet potatoes were selected as one of the foods tested for long-term space travel (Wilson and others 1998). Because of their high carotenoid content and good yields, OFSP have also been used in several small-scale studies to increase VA status (Haskell and others 2004, 2005; van Jaarsveld and others 2005; Low and others 2007a, 2007b). In this article, we evaluate the potential of sweet potatoes to prevent VA deficiency and determine the most important biochemical, physiological, and sociological factors influencing this potential.

VA and VA Deficiency

VA is an umbrella term for a family of compounds (retinol, retinoic acid, and retinyl esters (U.S. Inst. of Medicine [USIOM] 2000). The structure of beta-carotene and several important retinoids are shown below (Figure 1). Retinyl esters, the storage forms of VA, are represented by retinyl palmitate. Retinyl palmitate is the most common storage form of VA, but other common forms include retinyl stearate, retinyl linoleate, and retinyl myristate.

VA is an essential nutrient. Retinoids have numerous roles in genetic regulation (Blomhoff and Blomhoff 2006; McGrane 2007; Marceau and others 2007; Wolgemuth and Chung 2007; Mark

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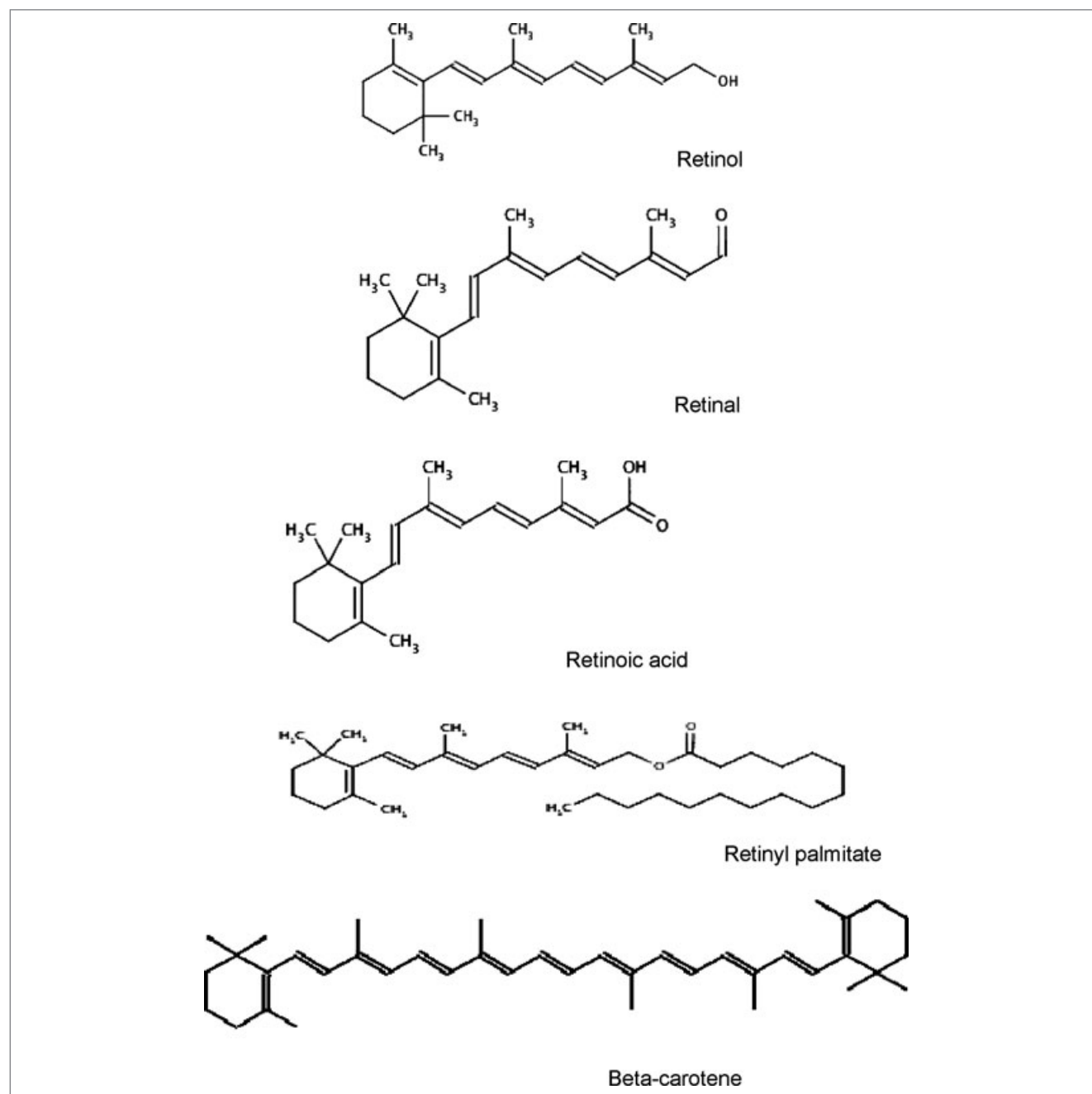


Figure 1—Representative structures of retinoids important for "Vitamin A" activity or metabolism.

and others 2009), the visual cycle (Blomhoff and Blomhoff 2006; Travis and others 2007; Ritter and others 2008), normal growth and development (Blomhoff and Blomhoff 2006; McGrane 2007; Marceau and others 2007; Wolgemuth and Chung 2007; Mark and others 2009), and immune function (Montrone and others 2009; Duriancik and others 2010).

VA deficiency is a serious health issue for much of the developing world. It is responsible for over 600000 deaths per year, mostly in young children or pregnant women (WHO 1995; West 2002; United Nations Childrens Fund [UNICEF] 2004; Black and others 2008; Table 1). VA deficiency is especially common in South-East Asia, and Sub-Saharan Africa, where 40% of preschool children are estimated to be deficient (West 2002; World Health

Organization [WHO] and Food and Agriculture Organization [FAO] 2004; WHO 2009). It is a leading cause of preventable blindness in the world (Sommer and others 1996; WHO and FAO 2004; WHO 2009). Providing VA to people with VA deficiency appears to decrease the severity of diseases such as influenza and diarrhea and reduces infant and maternal mortality by about 30% (Beaton and others 1993; Glasziou and Mackerras 1993; Fawzi and others 1993; West 2002; Sommer and Davidson 2002; Black and others 2008).

VA status is difficult to assess because retinoid metabolism is closely regulated, so that retinoid concentrations in blood, urine, and feces are not correlated with retinoid status except during severe deficiency or toxicity (Sommer and West 1996; Blomhoff

Table 1—Global estimates of vitamin A deficiency.

Reference	Year of survey	Nr with xerophthalmia (10 ⁶)			Nr with serum retinol <0.70 μmol/L (10 ⁶)		
		Preschool children	Pregnant women	All	Preschool	Pregnant	All
WHO 2009	2006	5.2	9.8	15.0	190	19.1	208.1
West 2002	2001	4.4	6.2	10.6	127	7.2	134.2
UNICEF 2004	2000			7.0			219
WHO 1995	1994			2.8			251

and Blomhoff 2006). Current methods to accurately measure VA status such as the modified dose-response test or methods using radio or stable isotopes are too expensive and labor intensive to be suitable for nutritional surveys (Lin and others 2000; Hickenbottom and others 2002; Burri and Clifford 2004; van Jaarsveld and others 2005; Ho and others 2009).

Thus, identifying marginal VA status and determining the onset of VA deficiency is not simple. Current guidelines estimate VA deficiency by determining the percentage of the population that have serum concentrations of retinol below 0.70 μmol/L, or by measuring the incidence of xerophthalmia, an eye condition that is characteristic of VA deficiency, but which is relatively rare (WHO 1995; West 2002; Sommer and Davidson 2002; United Nations Children's Fund (UNICEF) 2004; Black and others 2008).

The amount of VA needed from the diet depends on age, sex, and presumably on genetics and lifestyle. Recommended daily intakes of VA for healthy individuals called "dietary reference intakes," are shown in Table 2. The recommended nutrient intakes for VA are estimated by the World Health Organization (WHO and FAO 2004), and the U.S. Inst. of Medicine (USIOM 2000). Estimates differ between these organizations, especially for ado-

lescents and lactating women, because many of the estimates are based on extrapolations. For example, very few research studies have included pregnant or lactating women or adolescents in their populations.

VA deficiency prevention programs

VA deficiency is usually prevented by distributing high-dose VA supplements twice yearly (Pedro and others 2004; Aguayo and others 2005; Donnen and others 2007; Idindili and others 2007; Bezerra and others 2010). VA supplementation programs can be cost-effective nutrient interventions and, therefore, are supported by several national governments and international charitable organizations. However, programs have been difficult to sustain and outreach to poor rural populations can be problematic (Semba and others 2010). For example, India has supported national VA supplementation programs for 40 years, but they have attained less than 25% coverage, not reaching most poor rural populations (Stein and others 2006, supplemental material). Furthermore, there is risk that high-dose supplementation programs could cause toxicity in some infants, since some may receive more than one high-dose VA supplement (Mudur 2001).

Food fortifications have also been successful in preventing VA deficiency (Dary and Mora 2002; Kafwembe and others 2009; Oguntibeju and others 2009; Fiedler and Afidra 2010; Klemm and others 2010). Indeed, most developed countries fortify a variety of foods with VA. In the United States, most milk, dairy products, and some cereals are fortified with VA. However, food fortification can also be difficult to sustain, mostly because of the difficulties inherent in fortifying a food. The food must be consumed by almost everyone, including the poorest individuals. It must also be consumed with a narrow range of intakes: so that it prevents VA deficiency in most people, but does not cause toxicity in people who eat more than average amounts. A third strategy for preventing VA deficiency is to increase VA or VA-forming carotenoids in the diet.

VA dietary sources

People can get their VA preformed from animal source foods (Figure 1). The best sources are liver and organ meats and fish oils (United States Dept. of Agriculture [USDA] Agricultural Research Service [ARS] 2010). However, these foods are too expensive for most of the world's people to eat regularly. Fortunately, VA can also be formed from a variety of carotenoids. In low-income countries, about 82% of the total VA intake is from carotenoids in plants (van den Berg and others 2000; WHO 2009). Carotenoids are brightly colored phytonutrients found in fruits and vegetables. The most common VA-forming carotenoids are beta-carotene, alpha-carotene, and beta-cryptoxanthin (Britton 1995; USIOM 2000). Beta-carotene (Figure 1) is the most common pro-VA carotenoid in the food supply. It is found in many green- or orange-colored vegetables and fruits such as carrots, orange sweet potatoes, mangos, spinach, and pumpkin (Britton 1995; USDA ARS 2010).

Table 2—Recommended nutrient intakes for vitamin A.

Life stage	Recommended nutrient intake (WHO and FAO 2004) (μg RE/d)	Dietary reference intake (USIOM 2000) (μg RAE/d)
Infants 0 to 6 mo	375	400 (adequate intake; AI). No dietary reference intake has been established
Infants 7 to 12 mo	400	500 (AI)
Children 1 to 3 y	400	300
Children 4 to 6 y	450	400
Children 7 to 8 y	500	400
Children 9 y	500	600
Children 10 to 13 y	600	600
Adolescents 14 to 18 y	600	No separate values determined; 900 for males; 700 for females
Adult males 19 to 65 y	600	900
Adult males 65+ y	600	900
Adult females 19+ y (except when pregnant/lactating)	500	700
Adult females 65+ y	600	700
Pregnant females (14 to 18 y)	800	750
Pregnant females (19+ y)	800	770
Lactating females (14 to 18 years)	850	1200
Lactating females (19+ years)	850	1300

RE = retinol equivalents; 1 retinol equivalent = 6-μg β-carotene in food or 1-μg purified retinol. RE are the unit of measure used by the World Health Organization to describe the amount of vitamin A contributed to the diet by carotenoids.
RAE = retinol activity equivalents; 1 retinol activity equivalent = 12-μg β-carotene in food or 1-μg purified retinol. RAE are the unit of measure used by the USIOM to describe the amount of vitamin A contributed to the diet by carotenoids.

Conversion of Beta-Carotene to VA

In the test tube, 1 molecule of beta-carotene can be converted into 2 molecules of VA. VA is formed from carotenoids by simple one-step reactions via beta-carotene 15, 15'-monooxygenase (CMO1), which cleaves beta-carotene centrally to form 2 molecules of retinal. A secondary mechanism involves beta-carotene 9',10'-dioxygenase, CDO2, which cleaves beta-carotene, alpha-carotene, and beta-cryptoxanthin eccentrically to form 2 apocarotenals, the longer of which can then be oxidized to 1 molecule of retinal (Chichili and others 2005; Ho and others 2007; Biesalski and others 2007; Lietz and others 2010).

Carotenoids in the body are less effective. Isotopic dilution studies of beta-carotene conversion in healthy well-nourished subjects show variable conversion ratios, with some healthy volunteers forming negligible amounts of VA (Lin and others 2000; Hickenbottom and others 2002; Burri and Clifford 2004; Ho and others 2009). Currently, carotenoid conversion in the body is estimated to be 6- μ g beta-carotene: 1- μ g VA (WHO and FAO 2004) or 12- μ g beta-carotene: 1- μ g VA (USIOM 2000).

The reason for the relatively poor conversion of beta-carotene to VA is multifactorial. However, one important reason is that carotenoids are poorly absorbed from most foods (de Pee and others 1998; Veda and others 2006). Carotenoid absorption actually is highly variable and depends on the carotenoid, its food matrix, and the individual. Beta-carotene is better absorbed from orange-colored fruits and vegetables than from leafy green vegetables (de Pee and others 1998; O'Connell and others 2007). Beta-carotene absorption is much better when fed with oil (Huang and others 2000; Failla and others 2009; Bengtsson and others 2009) than without.

One human study has estimated that the VA equivalency of the carotenoids found in sweet potato. The VA equivalency of beta-carotene fed to Bangladeshi men with moderate VA stores was estimated to be 13.4- μ g beta-carotene to 1- μ g retinol (Haskell and others 2004).

However, people and animal models with low VA status appear to convert a greater percentage of beta-carotene to VA (Ribera-Mercado and others 2000; Howe and Tanumihardjo 2006; Porter-Dosti and others 2006; Tanumihardjo 2008). For example, the conversion ratio of beta-carotene to VA in poorly nourished Filipino children varied inversely with VA status (Ribaya-Mercado and others 2000). Similarly, carotenoid conversion in an appropriate small animal model (Lee and others 1999) for beta-carotene metabolism (the Mongolian gerbil) showed a highly negative correlation ($R = 0.88$) with liver reserves (Tanumihardjo 2008). The conversion ratio in VA depleted gerbils fed a maize meal was 3- μ g beta-carotene to 1- μ g retinol (Howe and Tanumihardjo 2006).

Food-Based Interventions for Improving VA Status

Growing fruits and vegetables that are rich in VA-forming carotenoids is a good alternative to providing VA supplements or fortifying foods (Ruel 2001; Faber and van Jaarsveld 2007; Tanumihardjo 2008; Tanumihardjo and others 2008; Loechl and others 2009; Bouis and Welch 2010). Fruits and vegetables can provide a variety of nutrients in addition to VA, and they can provide income to small farmers and shopkeepers whose families are at risk for nutrient deficiencies. Long-term sustainability of food-based programs might be achieved, because fruit and vegetable seeds can be harvested and shared at a local level, instead of being provided by a national program.

A variety of foods naturally rich in beta-carotene have been used successfully in small-scale interventions to increase VA status (de Pee and others 1998; Haskell and others 2004, 2005; van Jaarsveld and others 2005; Low and others 2007a, 2007b; Burri and Turner 2009). In addition, a variety of staple foods, which are not normally good sources of beta-carotene, are being "biofortified" with beta-carotene through genetic engineering or selective breeding. These foods include maize, cassava, and rice (Thakkar and others 2007; Davis and others 2008; Naqvi and others 2009; Vallabhaneni and others 2009; Yan and others 2010).

OFSP are well suited for nutrient interventions because they are a naturally rich source of beta-carotene. They have been used to increase VA status in several small-scale interventions (Haskell and others 2004, 2005; van Jaarsveld and others 2005; Low and others 2007a, 2007b; Burri and Turner 2009; Howe and others 2009; Zeng and others 2009). In addition, they are already a secondary staple food in much of Africa and Asia (USDA Economic Research Service [ERS] 2009). Therefore, programs to improve and distribute OFSP are being tested by multinational nutritional programs such as HarvestPlus for their potential to improve VA status on a national level. These trials were uniformly successful, with as little as $\frac{1}{2}$ cup/d improving VA reserves in children (van Jaarsveld and others 2005).

Carotenoid Content of Sweet Potatoes

Sweet potatoes vary in color and carotenoid concentration. The primary VA-forming carotenoid in sweet potatoes is beta-carotene (Bengtsson and others 2008; Wu and others 2008; USDA ARS 2010), although small amounts of alpha-carotene and beta-cryptoxanthin can be found in some varieties. The concentration of beta-carotene depends largely on the variety of sweet potato (Hangenimana and others 1999a; Hangenimana and Low 2000; Kidmose and others 2006, 2007, 2009; Bengtsson and others 2008; Wu and others 2008; USDA ARS 2010). There is a very wide (1100-fold) range of beta-carotene concentrations in sweet potato (Table 3).

Essentially, carotenoid concentrations varied with sweet potato color. The more orange the color the higher the carotenoid content (Ameny and Wilson 1997; Takahata and others 1993). Thus, white-fleshed sweet potatoes < cream < yellow = purple < light orange < orange (Table 3).

Beta-carotene concentrations also vary with growing, harvesting, and storage conditions (Bengtsson and others 2008, Hagenimana and others 1999a), farming site (K'osambo and others 1998), season (Liu and others 2009), root age (K'osambo and others 1998; Hagenimana and others 1999b), drought (van Heerden and Laurie 2008), and virus infestation (Kapinga and others 2009).

Effects of cooking and storage on carotenoid content of sweet potatoes

There are a great variety of cooking and processing methods for sweet potatoes (Woolfe 1992; Emenhiser and others 1999; Kudoh and Matsuda 2000; Rodriguez-Amaya 2003; Sulaeman and others 2003; Mohapatra and others 2007; Low and van Jaarsveld 2008). Almost all sweet potatoes are eaten cooked, although they can also be fermented into alcohol (Ray and Sivakumar 2009) and processed into flour (Ahmed and others 2010), buds (Valdez and others 2001), curd (Mohapatra and others 2007), or yogurt (Kudoh and Matsuda 2000).

Carotenoids are well retained with most cooking methods (Table 4). Carotenoid losses are minimal, with typical losses of 0% to 12% (Chandler and Schwartz 1988; Rodriguez-Amaya 1997;

Table 3—Estimated amount of carotenoids in 100 g (edible portion) of sweet potato varieties.

Reference	Sweet potato varieties with lowest and highest carotenoid concentrations, listed sequentially (<i>n</i> = number of varieties tested)	Estimated beta-carotene content ($\mu\text{g}/100\text{ g}$)
Lako and others 2007	<i>n</i> = 4, Honiara to orange var	ND to 15000
Kidmose and others 2007	<i>n</i> = 6, Odiewo (white) to Tainung (orange)	1240 to 10800
Wu and others 2008	<i>n</i> = 14, Nanzi 8 (purple) to Xushu 22.5 (orange)	5320 to 8400
Bengtsson and others 2008	<i>n</i> = 7, SP 004/1 to SPK 004/6	3120 to 10244 (calculated from dry matter)
Failla and others 2009	<i>n</i> = 10, 3163420269 to 1900094.2	11120 to 28100
USDA 2009	Not identified	5501 to 11509
Takahata and others 1993		
Yellow-white	<i>n</i> = 3, Beniazuma to Kagenasama	ND
Orange	<i>n</i> = 20, PI208886 to SPV-61	7400 to 18700
Fonseca and others 2008		
Cream	<i>n</i> = 1, Rosinha de Verdan	437
Orange	<i>n</i> = 1, IAPAR 69	10120
Teow and others 2007		
White	<i>n</i> = 1, Xushu 18	20
Light purple	<i>n</i> = 2, 12.9 to 13.6	2230 to 5660
Purple	<i>n</i> = 4, 13.18 to 12.5	540 to 4690
Yellow	<i>n</i> = 2, 11.12 to 13.1	150 to 230
Light orange	<i>n</i> = 3, 12.17 to 12.7	1180 to 2980
Orange	<i>n</i> = 7, 13.14 to 11.20	4490 to 22600
Hangenimana and others 1999a (beta-carotene)		
White	<i>n</i> = 9, Ex-Diana, CIP440066	Trace to 19.6
Cream-white	<i>n</i> = 1, Xiang shu 6	274
Cream	<i>n</i> = 6, CIP440160 to CIP187004.1	Trace to 1071
White-yellow	<i>n</i> = 4, KEM20 to CIP440023	Trace to 37.4
Cream-yellow	<i>n</i> = 3, CARI9 to K148	174 to 473
Cream-purple	<i>n</i> = 1, Estrella	486
Purple-white	<i>n</i> = 2, Morodomaravi to Capadito	162 to 249
Purple-cream	<i>n</i> = 1, Cascajo morado	199
Yellow-purple	<i>n</i> = 1, IRA502	Trace
Light yellow	<i>n</i> = 3, Mabrouka to Xushu18	49.8 to 112
Yellow	<i>n</i> = 1, KEMB 10	627
Light orange	<i>n</i> = 3, Maria Angola to Mamala	111 to 2217
Orange	<i>n</i> = 7, Zapallo to TIB11	4085 to 7984

ND = not detectable.

K'osambo and others 1998; van Jaarsveld and others 2006; Bengtsson and others 2008, 2009; Failla and others 2009; USDA ARS 2010; Ahmed and others 2010). Dehydration causes greater losses, of up to 41% (Fonseca and others 2008).

Sweet potatoes store well. They retain most of their carotenoids for at least 50 d (Watanabe and others 1999) or longer (Chattopadhyay and others 2006). Thus, the effects of cooking and storage method are relatively negligible, causing losses of about 0% to 20%.

Bioaccessibility of sweet potatoes

Although OFSP contain large amounts of beta-carotene, not all of it is accessible. Carotenoid bioaccessibility is defined as the fraction of carotenoids transferred by food to mixed micelles, therefore becoming accessible for subsequent uptake by the intestinal mucosa. Carotenoid bioaccessibility depends on the food matrix, the

Table 4—Effect of processing on carotenoids in 100 g (edible portion).

Variety/treatment	Estimated carotenoid content ($\mu\text{g}/100\text{ g}$)	Reference
Tainung		
Raw	10800	Kidmose and others
Boiled	9200	2007 (all- <i>trans</i> beta-carotene)
Roasted	7910	
SPK 004		
Raw	6200	Kidmose and others
Boiled	7830	2007 (all- <i>trans</i> beta-carotene)
Roasted	4630	
Salyboro		
Raw	4840	Kidmose and others
Boiled	4270	2007 (all- <i>trans</i> beta-carotene)
Roasted	5140	
Zapallo		
Raw	3480	Kidmose and others
Boiled	3570	2007 (all- <i>trans</i> beta-carotene)
Roasted	4140	
199062.1		
Raw	2010	Kidmose and others
Boiled	1610	2007 (all- <i>trans</i> beta-carotene)
Roasted	1770	
Nyathi-Odiwo		
Raw	1240	Kidmose and others
Boiled	560	2007 (all- <i>trans</i> beta-carotene)
Roasted	610	
Ejumula		
Uncooked (DW)	10810	Bengtsson and others
Boiled	7300	2008
Steamed	7370	
Deep fried	10310	
SPK004/6		
Uncooked (DW)	31450	Bengtsson and others
Boiled	25260	2008
Steamed	24990	
Deep fried	25310	
(NI)		
Cylinders		
Boiled (FW)	57060	Bengtsson and others
		2009
Steamed	68240	
Microwaved	92940	
(NI)		
Slices		
Blanched/boiled	91520	
Boiled	86690	
(NI)		
Canned, drained	5501	USDA 2009
Boiled	9444	
Baked	11509	

(NI) = variety not indentified; DW = dry weight; FW = fresh weight.

type of fiber and fat in the food, and the heat and homogenization caused by food processing (Veda and others 2006; Bengtsson and others 2009; Tumuhimbise and others 2009). Specifically, the extent of carotenoid bioavailability from sweet potatoes depends on the sweet potato variety and cooking and processing methods. Bioaccessibility varied with processing method so that raw < baked < steamed/boiled < deep fried (Bengtsson and others 2009; Tumuhimbise and others 2009).

In all studies, beta-carotene bioaccessibility increased greatly with fat, as is typical with other foods. One simulated digestion showed that only 0.6% to 3% of sweet potato carotenoid was micellized, increasing to 7% in highly processed baby food (Failla and others 2009). However, a 2nd study (Bengtsson and others 2009) showed much higher bioaccessibility. The accessible beta-carotene in the micellar phase varied from 0.5% to 1.1% without

fat, increasing to 11% to 22% with 2.5% fat. The percentage of accessible beta-carotene in the supernatant phase was higher, between 24% to 41% without fat and 28% to 46% with added fat. Furthermore, a study in VA-depleted Mongolian gerbils fed OFSP with 3%, 6%, and 12% fat for 3 wk showed that carotenoid absorption increased as the amount of fat in the diet increased (Mills and others 2009). All OFSP diets maintained VA status in gerbils, while the higher fat (12% fat) diets improved status. This study also showed that stir-frying doubled the efficiency of beta-carotene incorporation into micelles. Finally, a human study, feeding sweet potatoes with fat, calculated bioavailability as 65% for beta-carotene beadlets and 37% for sweet potatoes (Huang and others 2000).

Thus, the bioaccessibility of beta-carotene from sweet potato can be very low (<1%) if fed without fat. Even a small amount of fat appears to increase beta-carotene bioaccessibility in sweet potatoes by 2- to 20-fold. Even so, only about 25% (11% to 48%) of the beta-carotene in sweet potatoes is bioaccessible, and thus available to be absorbed into the intestine. Note that the fraction of bioaccessible beta-carotene (25%) is similar to the conversion ratio for beta-carotene to VA estimated for VA-deficient gerbils (Howe and Tanumihardjo 2006) and people (Tanumihardjo 2008), which is 33%. This is reasonable, since carotenoid metabolism studies (Lin and others 2000; Burri and others 2001; Hickenbottom and others 2002; Ho and others 2009) suggest that poor absorption of carotenoids from food is the major reason for the low conversion ratio of beta-carotene to VA.

Consumer Preference

Although many sweet potato growing countries traditionally eat cream or white sweet potatoes, studies that have asked them to switch from white to orange sweet potatoes have found little resistance (Low and others 2007b, 2008; Naico and Lusk 2010). Thus, the impact of consumer preference on the success of OFSP programs to prevent VA deficiency should be small, perhaps on the order of 5% to 10%.

The amount of OFSP needed to supply VA to 1 person

We evaluated sweet potatoes for their potential to prevent VA deficiency on a worldwide basis. First, we found the beta-carotene, alpha-carotene, and beta-cryptoxanthin concentrations of sweet potatoes (Table 2). Second, we reviewed the data on the effects of cooking and processing (Table 3) and carotenoid bioaccessibility from sweet potato. After adjusting the carotenoid concentrations in OFSP for losses due to cooking, storage, and poor bioaccessibility, we calculated the amounts of OFSP needed to meet the VA requirement for 1 person at risk, at different life stages (Table 1). We calculated bioaccessible beta-carotene in OFSP with the following equation:

$$\begin{aligned} &\text{Bioaccessible beta-carotene in OFSP} \\ &= \text{Beta-carotene in OFSP} \\ &\quad \times \text{Fraction retained after cooking and storage} \\ &\quad \times \text{Bioaccessible fraction} \end{aligned}$$

When beta-carotene in OFSP = 4085 to 22900 $\mu\text{g}/100\text{ g}$ (Takahata and others 1993; Hangenimana and others 1999; Teow and others 2007); fraction retained after cooking and storage = 0.90 and bioaccessible fraction = 0.25.

Therefore, the concentrations of bioaccessible beta-carotene in OFSP range from 919 to 5152 $\mu\text{g}/100\text{ g}$ OFSP; or 9- to 52- μg bioaccessible beta-carotene/g OFSP.

We calculated the amount of OFSP it would take to supply 1 person with 100% of the VA needed, using data from Table 1 and 2. The conversion ratio we used was dependent on the VA status of the individual. Well-nourished individuals, with good VA status, convert less beta-carotene to VA than poorly nourished people with low VA status (Ribaya-Mercado and others 2000; Howe and Tanumihardjo 2006; Tanumihardjo 2008). We estimated that women and children with good VA status had a retinol equivalency ratio of 12- μg beta-carotene: 1- μg retinol (USIOM 2000). Poorly nourished women and children are likely to have a smaller retinol equivalency ratio of perhaps 3- μg beta-carotene: 1- μg retinol, which depends mainly on carotenoid bioaccessibility (Tanumihardjo 2008).

We used these concentrations of bioaccessible beta-carotene, and the weight of 1 cup of sweet potato (USDA ARS 2010), to calculate the amount (in grams and in cups/day) of OFSP needed to supply the VA requirements of 1 person, at different life stages (Table 1). These amounts should correspond to the amounts of OFSP needed by representative individuals with marginal VA status.

In addition, we used the beta-carotene concentrations of OFSP (of 4085 to 22900 $\mu\text{g}/100\text{ g}$) and the more conservative conversion ratio of 12- μg beta-carotene: 1- μg retinol for well-nourished adults (USIOM 2000).

Therefore, the grams/day of OFSP needed to meet the requirements for 1 person with marginal VA deficiency is calculated as:

$$\begin{aligned} &\text{Grams per day OFSP (during marginal vitamin A deficiency)} \\ &= ((\mu\text{ gRE/day})/(\mu\text{ g bioaccessible beta-carotene/} \\ &\quad \text{gram sweet potato})) \end{aligned}$$

For a 10- to 13-y old with marginal deficiency this ranges from:

$$\begin{aligned} \text{Grams per day OFSP} &= 600(\mu\text{g RE/d})/51.5\mu\text{g bioaccessible} \\ &\quad \text{beta-carotene/g sweet potato} = 12\text{ g/d;} \end{aligned}$$

to

$$\begin{aligned} \text{Grams per day OFSP} &= 600(\mu\text{g RE/d})/9.2\mu\text{g bioaccessible} \\ &\quad \text{beta-carotene/g sweet potato} = \text{or } 66\text{ g/d.} \end{aligned}$$

The grams/day of OFSP needed to meet the requirements for 1 person with good VA status is calculated as:

$$\begin{aligned} &\text{Grams per day OFSP (good vitamin A status)} \\ &= ((\mu\text{g RE/day})/((\mu\text{g beta-carotene/gram sweet potato}) \\ &\quad \times (\text{conversion ratio } 12/1))) \end{aligned}$$

For a 10- to 13-y old with good status this ranges from:

$$\begin{aligned} \text{Grams per day OFSP} &= 600(\mu\text{g RE/d})/(229\mu\text{g} \\ &\quad \text{beta-carotene/g sweet potato}/12) = 31\text{g OFSP/d;} \end{aligned}$$

to

$$\begin{aligned} \text{Grams per day OFSP} &= 600(\mu\text{g RE/d})/(40.8\mu\text{g} \\ &\quad \text{beta-carotene/g sweet potato}/12) = 176\text{g OFSP/d.} \end{aligned}$$

The ranges of OFSP/day needed to meet VA requirements for well nourished and marginally VA-deficient individuals are shown in Figure 2. We also calculated the amounts of OFSP/day in terms of cups/day. This allows one to determine if the portion size could reasonably be fed as part of a normal diet. One cup of OFSP was

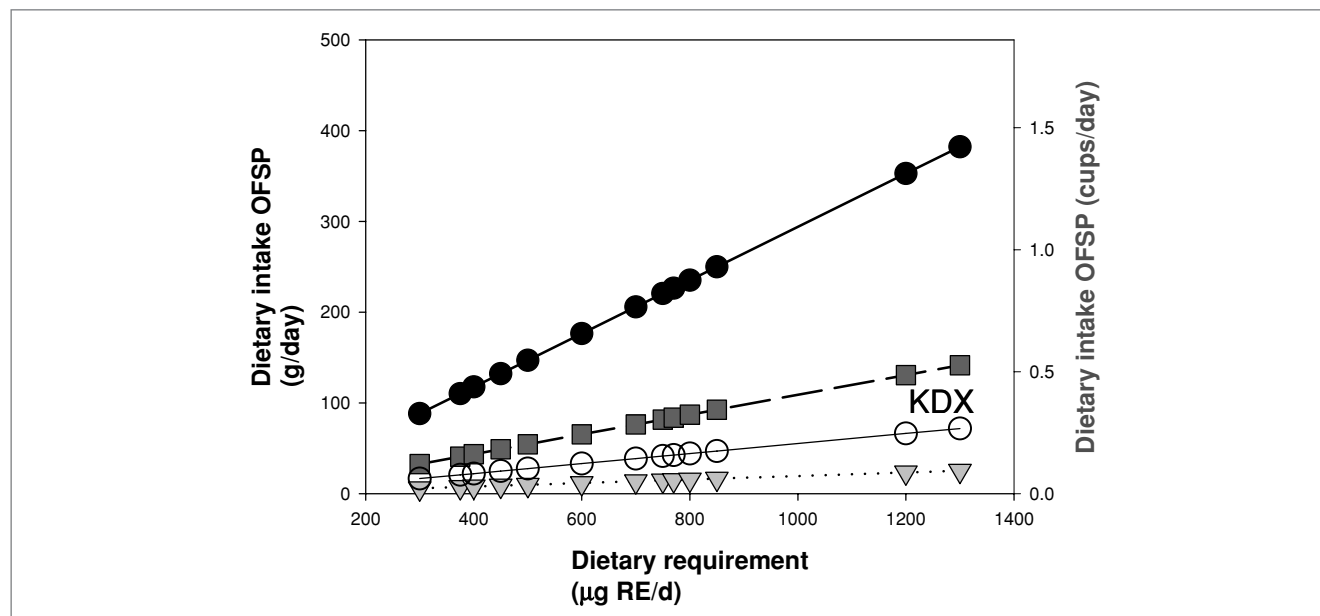


Figure 2—Estimated amounts (gram/day and cups/day) needed to meet the recommended dietary intakes at different life stages. ● = High estimate of OFSP intake needed to meet the dietary requirement of vitamin A for vitamin A-replete individuals. ■ = High estimate of OFSP intake needed to meet the dietary requirement of vitamin A for individuals with marginal vitamin A status. ○ = Low estimate of OFSP intake needed to meet the dietary requirement of vitamin A for vitamin A-replete individuals. ▽ = Low estimate of OFSP intake needed to meet the dietary requirement of vitamin A for individuals with marginal vitamin A status.

estimated to weigh 255 g (USDA ARS 2010). Therefore, to calculate the cups/day of OFSP that would be needed to supply 100% of the requirement for VA, one must divide the gram OFSP/day by 255 g/cup.

Thus, the 10- to 13-y old in the examples above would need 0.05 to 0.26 cups/day if he had marginal VA status, or 0.12 to 0.69 cups/d if she had adequate VA status. The daily intakes of OFSP, in cups/day, needed to meet 100% of the requirements for VA, are also shown in Figure 2. These results show that the daily intake of OFSP that would supply 100% of the VA requirement is reasonable for all populations, with the possible exception of lactating women, who would have to eat as much as 1.5 cups/d.

The amount of OFSP required to supply the population most at risk for VA deficiency

The amount (in metric tons) of OFSP needed to supply VA to all the 208 million people most at risk for VA deficiency was calculated and compared to the amount of sweet potatoes grown per year. The people most at risk for VA deficiency are the 190 million preschool children and 19.1 million pregnant women from low-income, food-deficit countries estimated to have low VA status (Table 1, WHO 2009). We assumed that 75% of the preschool children were aged 1 to 3 y (with a requirement of 400 RE/d), and 25% were aged 4 to 5 (with a requirement of 450 RE/d). Pregnant women had a requirement of 800 RE/d. We assumed that 20% of these pregnant women were also lactating, with the higher requirement of up to 1300 RAE/d). Since the estimate is for the people most at risk for VA deficiency, we are assuming these people have marginal VA status.

Therefore, 147.5 million children have an estimated requirement of 400 RE/d; and each needs 23.4 to 131.2 g OFSP/d.

A total of 47.5 million children have an estimated requirement of 450 RE/d; and each needs 26.3 to 147.6 g OFSP/d.

A total of 15.28 million pregnant women have an estimated requirement of 800 RE/d; and each needs 46.8 to 262.5 g OFSP/d.

A total of 3.82 million pregnant women are also lactating and have a requirement of 1300 RE/d and each needs 76.1 to 426.5 g OFSP/d.

Therefore, the smallest amount of OFSP needed for the 208.1 million people most at risk is:

$$\begin{aligned}
 & (147500000 \times 23.4 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & + (47500000 \times 26.3 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & + (15280000 \times 46.8 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & + (3820000 \times 76.1 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & = 2.083 \times 10^{12} \text{ g OFSP/y, or 2.083 million metric tons/y.}
 \end{aligned}$$

The higher and probably more realistic amount of OFSP needed for these 208.1 million people is:

$$\begin{aligned}
 & (147500000 \times 131.2 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & + (47500000 \times 147.6 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & + (15280000 \times 262.5 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & + (3820000 \times 426.5 \text{ g OFSP/d} \times 365 \text{ d/y}) \\
 & = 11.681 \times 10^{12} \text{ g OFSP/y, or 11.681 million metric tons/y.}
 \end{aligned}$$

Production, Yield, and Areas Harvested with Sweet Potato

It would be best to compare these amounts to the current production of OFSP. Unfortunately, there is little information on this production. We searched the internet and reference databases (PubMed and Agricola) between July and September 2010. Key words used were “caroten*,” “RN = 7235-40-7,” “RN = 7488-99-5,” and “RN = 472-70-8.” Keywords for sweet potato were “sweet potato” and “*Ipomoea batatas*.” Key words for production were “production,” “yield,” and “harvest.” Key words for sweet potato color were “color,” “hue,” “orange,” “cream,” “white,”

Table 5—Production quantity, area harvested, and yield for sweet potatoes (FAOSTAT 2009).

	Production metric tons	Yield metric tons/hectare	Area harvested hectares
World	106501450	127406	8359185
Asia	87797235	194026	4525016
Africa	15025235	44191	3401060
Low-income, food-deficit countries	100878482	128124	7873472
USA	836560	212454	39376
China	80522926	213009	3780254
Nigeria	3318000	30000	1106000
Uganda	2707000	45191	599000
Indonesia	1876944	107742	174205
Vietnam	1323900	81621	162200

“yellow,” and “purple.” The articles retrieved were hand searched to retrieve other relevant articles. The searches were restricted to articles in English. We found essentially no international or national data on OFSP (USDA ERS 2009). The U.S. data are of general sweet potato production, which includes both OFSP and white, cream, yellow, and purple sweet potatoes that contain low concentrations of beta-carotene. Thus, the production data available probably overestimate OFSP production by a factor of 2 or 3. With this caveat, the major producers of sweet potatoes, their production volume, and emerging issues that might influence their ability to increase production were identified. Production values were used to estimate whether current production of sweet potatoes is sufficient to supply the people most at risk for V deficiency with 100% of their VA requirement.

The production, yield, and land area harvested for the world, Asia, Africa, low-income countries with food deficits, and the United States are shown in Table 5. The most current information for these food production figures is from 2007. Current production is believed to be similar. Table 5 shows current world production of sweet potatoes is 106.5 million metric tons, which is much higher than the 2.08 to 11.68 million metric tons that would be required to supply 100% of the VA for the people most at risk for VA deficiency in the world.

Furthermore, unlike many crops, most sweet potatoes are produced by low-income, food-deficit countries. China is the major producer of sweet potatoes in the world, producing about 80% of the crop. Other major producers are Nigeria, Uganda, Indonesia, and Vietnam. Yield is quite high in China but is much lower in African countries. If people in Africa could use the expertise of China's agriculture to increase their sweet potato yields to their current levels, then Nigeria and Uganda would produce 23558795 and 12759517 metric tons of sweet potatoes, respectively, on the same amount of land they are using now.

Of course, the availability and acceptability of a food is also an important indicator of how well it might serve as a food-based intervention to prevent VA deficiency. A review of the extent and distribution of sweet potato production suggests that sweet potatoes are available in and accepted among many populations with VA deficiency.

Data on the availability of foods for consumption are less recent than food production and are subject to even more uncertainty. These data are presented in Table 6 and show relatively low values. Even so, the estimates of consumption of sweet potatoes in low-income, food-deficit countries would be sufficient to provide 100% of the VA required by preschool children, if the sweet potatoes available were OFSP with high beta-carotene concentrations.

Table 6—Availability for consumption of sweet potatoes (2003 data; FAOSTAT 2009).

Area	Sweet potatoes (gram/person/day)
World	27.40
Asia	38.36
Africa	30.14
Low-income, food-deficit countries	41.10
USA	5.48

FAOSTAT. <http://www.fao.org/economic/ess/publications-studies/statistical-yearbook/en/html>

Despite this encouraging data, most of the countries that are major producers of sweet potatoes (China, Nigeria, and Uganda) also have moderate levels of VA deficiency (WHO 1995; West 2002; UNICEF 2004; Black and others 2008). This is rather surprising, since sweet potatoes are a popular food, and new varieties of sweet potatoes appear to be accepted easily (Laurie and Magoro 2008). Consumers also enjoy new products made with sweet potato flour (Low and other 2008; Naico and others 2010).

Environmental Impact

Increasing sweet potato production could increase its availability as a source of VA. However, producing large quantities of any single crop, even for nutritional and humanitarian purposes, is not a trivial undertaking (Ruel 2001; Amede and others 2004; Bovell-Benjamin 2007). It has impact on a variety of societal, economic, and health issues. This impact is not always beneficial. For example, red palm oil has been very effective against VA deficiency in small-scale trials and could be a highly effective food-based intervention to prevent VA deficiency (Zeba and others 2006; Oguntibeju and others 2009; Burri and Turner 2009). However, with the exception of Burkino-Faso, it has not been tested for this purpose on a national level. Even Malaysia and Indonesia, which currently produce most of the world's red palm oil, do not use it at a national level, nor is there evidence that its consumption has decreased VA deficiency (USDA ERS 2009; Burri and Turner 2009). In fact, most red palm oil is stripped of carotenoids, than used as biofuel (Thoenes 2006; Greenpeace 2007). Furthermore, the production of biofuels in these countries has been associated with severe environmental degradation (Greenpeace 2007; Burri and Turner 2009).

Currently, there is little evidence that sweet potato production is associated with especially unfavorable environmental impact (Wood 2002; Bovell-Benjamin 2007; Burri and Turner 2009). However, sweet potatoes, such as red palm oil and maize, are used for animal feed, fish feed (Dongmeza and others 2009), and diverted for biofuel production (Comis 2008). Currently, half of the sweet potatoes grown in Asia are used in animal feed (Consultative Group on International Agriculture Research [CGIAR] 2009). Thus, production increases may not lead to a more nutritious food supply.

Relative Importance of Factors Influencing the Effectiveness of Sweet Potatoes for Preventing VA Deficiency

Several factors might be important variables for determining the effectiveness of sweet potatoes as a food-based intervention to prevent VA deficiency. Most of these factors influence the carotenoid concentration of the sweet potato. These factors include variety of sweet potato (Table 3) and growing, harvesting, and storage conditions (Chandler and Schwartz 1988; K'osambo

and others 1998; Hagenimana and others 1999a, 1999b; Bengtsson and others 2008; van Heerden and others 2008; Kapinga and others 2009; Liu and others 2009), and cooking method and conditions (Rodriguez-Amaya 1997; K'osambo and others 1998; van Jaarsveld and others 2006; Bengtsson and others 2008; 2009;

Failla and others 2009; USDA 2009; Ahmed and others 2010). In addition, the food matrix and the presence or absence of fat influence carotenoid bioaccessibility (Huang and others 2000; Howe and Tanumihardjo 2006; Veda and others 2006; Tumuhimbise and others 2009; Bengtsson and others 2009; Failla and others 2009;

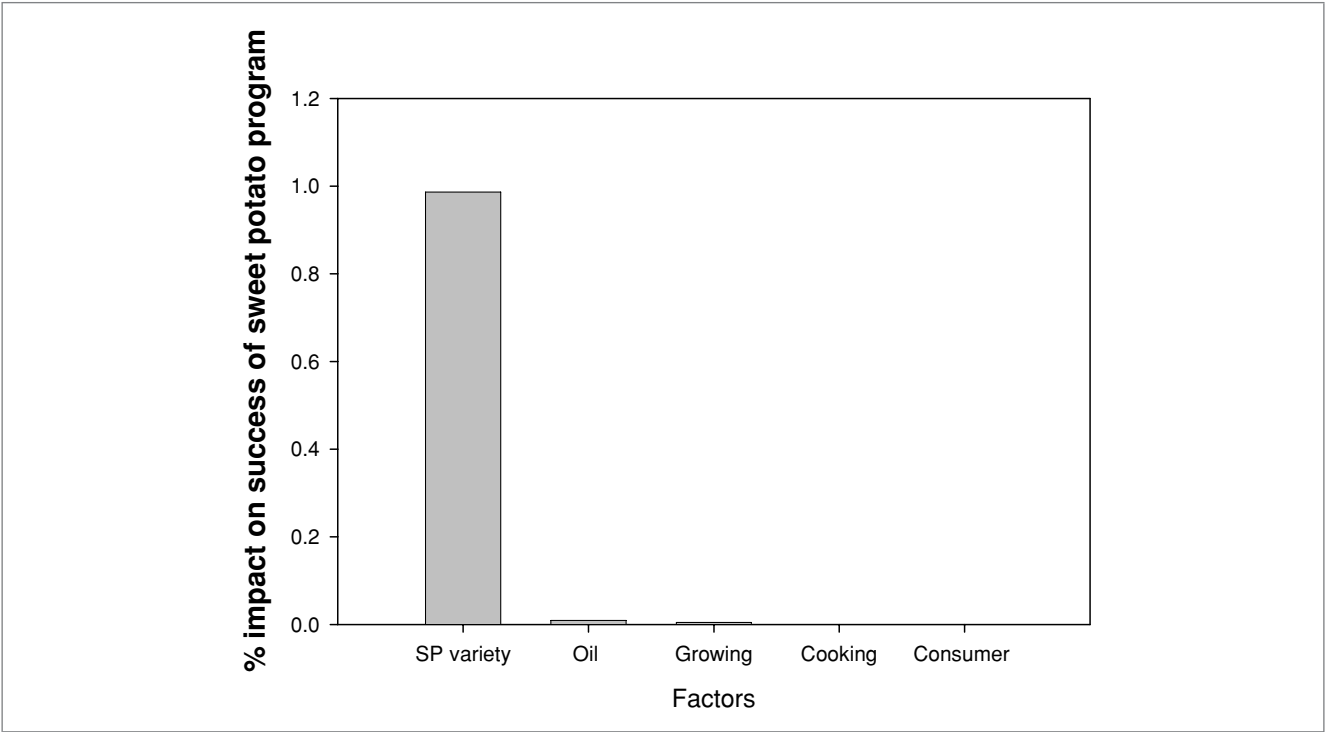


Figure 3—Factors influencing success of sweet potato treatments for improving vitamin A status. SP variety = sweet potato variety; Oil = prepared with or without cooking oil; Growing = growing and harvesting conditions; Cooking = cooking and storage conditions; Consumer = consumer preference.

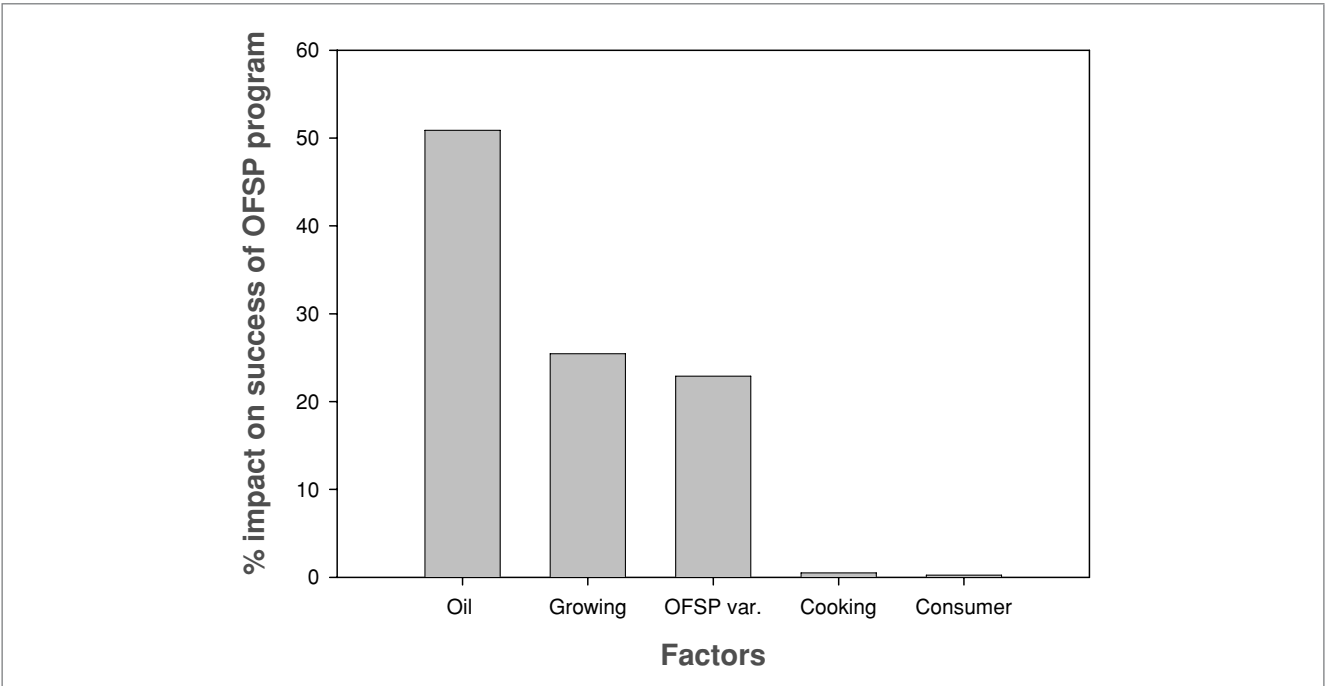


Figure 4—Factors influencing success of OFSP treatments for improving vitamin A status. Oil = prepared with or without cooking oil; Growing = growing and harvesting conditions; OFSP variety = orange-fleshed sweet potato variety; Cooking = cooking and storage conditions; Consumer = consumer preference.

Mills and others 2009). We estimated the relative importance of these factors by comparing the ranges of values reported for each variable.

The effect of sweet potato variety on carotenoid concentrations (Table 2) is very great, ranging from negligible to $\geq 22000 \mu\text{g}/100 \text{ g}$, a variation of at least 110000%. This would account for about 98.6% of the variability in the effectiveness of sweet potato interventions. No other factor approaches this impact. Thus, the best way of influencing the effectiveness of sweet potatoes as an intervention for preventing VA deficiency is to substitute white, cream, yellow, purple, or light orange sweet potatoes with OFSP.

Even among OFSP, the range is at least from 4000 to 22000 $\mu\text{g}/100 \text{ g}$, or 450%. The effect of adding oil to the OFSP preparation, or the diet is somewhat greater, ranging from 200% to 2000%. Growing and harvesting conditions, barring natural disaster or negligence, have similar impacts, or about 500%. The effects of cooking and storage techniques (other than adding oil) are much smaller with a range of 0% to 20%. The effects of consumer preference may be even smaller, at 0% to 10%. We used these data to construct Pareto charts, which rank the factors influencing the likelihood of success of sweet potato interventions to improve VA status in terms of their importance (Figure 3 and 4). The magnitude of each factor depends on the amount of variation it can cause in carotenoid concentration. Thus, the impact of sweet potato variety, growing conditions, and the variety of OFSP are 110000, 500, and 450, respectively. We used midpoint values to estimate the magnitude of the effects of adding oil to cooking, other cooking and storage techniques, and consumer preference. Thus, we estimated the magnitude of these factors to be 1000, 10, and 5, respectively. Figure 3 clearly demonstrates that the most important variable determining the effectiveness of sweet potatoes for preventing VA deficiency is the variety of the sweet potato.

If only OFSP are considered, then the determining factors (as shown in Figure 4) are the presence of fat in the food preparation > growing and harvesting conditions = variety of OFSP > cooking and storage > consumer preference.

Conclusion

Many common varieties of OFSP are excellent sources of VA. They are relatively simple to grow, durable, and are easy to prepare. Currently, just over half of the sweet potatoes grown are eaten by humans. Therefore, OFSP have considerable potential as a nutritious, sustainable source for VA. This potential could be increased substantially, if farmers in developing countries replaced white, cream, yellow, and purple sweet potatoes with OFSP. Further improvements could result from cooking OFSP with oil, and selecting varieties of OFSP with high carotenoid concentrations. In addition, the potential of OFSP to prevent VA deficiency would be increased if farmers in Africa could increase their yield to that obtained in China. Translating small-scale interventions to national and international programs that prevent vitamin deficiencies and improve health around the world will require planning, work, and a good appreciation of the economic and environmental issues involved in growing and distributing large quantities of crops for food-based interventions. However, it appears that increasing the amount of OFSP available to populations at risk for VA deficiency may result in good, sustainable, food-based interventions for preventing this nutritional disease.

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